23/01/2025
Evening

Aakash
Medical | IIT-JEE | Foundations

Corporate Office : AESL, 3rd Floor, Incuspaze Campus-2, Plot-13, Sector-18, Udyog Vihar, Gurugram, Haryana-122018

Answers & Solutions

Time : 3 hrs. *for* M.M. : 300

JEE (Main)-2025 Phase-1

[Computer Based Test (CBT) mode]

(Mathematics, Physics and Chemistry)

IMPORTANT INSTRUCTIONS:

(1) The test is of **3 hours** duration.

(2) This test paper consists of 75 questions. Each subject (MPC) has 25 questions. The maximum marks are 300.

(3) This question paper contains **Three** Parts. **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is **Mathematics**. Each part has only two sections: **Section-A** and **Section-B**.

(4) **Section - A :** Attempt all questions.

(5) **Section - B :** Attempt all questions.

(6) **Section - A (01 – 20)** contains 20 multiple choice questions which have **only one correct answer**. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.

(7) **Section - B (21 – 25) contains** 5 **Numerical value** based questions. The answer to each question should be rounded off to the **nearest integer**. Each question carries **+4 marks** for correct answer and **–1 mark** for wrong answer.



MATHEMATICS

SECTION - A

Multiple Choice Questions: This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Choose the correct answer:

1. If the area of the region

$\{(x,y): -1 \le x \le 1, 0 \le y \le a + e^{|x|} - e^{-x}, a > 0\}$ is

$\frac{e^2+8e+1}{e}$, then the value of a is:

(1) 5 (2) 7

(3) 8 (4) 6

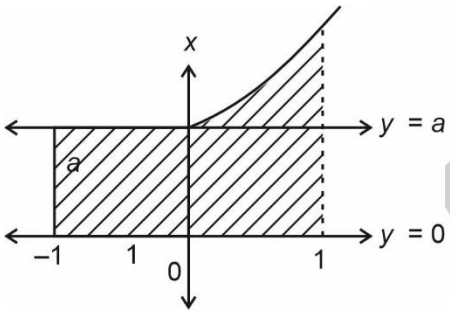
Answer (1)

Sol. $y \in \left[0, a + e^{|x|} - e^{-x}\right]$

(i) If $x \ge 0 \Rightarrow y \in \left(0, a + e^{x} - \frac{1}{e^x}\right)$

if $x < 0 \Rightarrow y \in \left(0, a + e^{-x} - e^{-x}\right)$

$\Rightarrow \quad y \in (0, a)$



$$\text{Area} = (a) + \int_0^1 (a + e^x - e^{-x})\,dx = \frac{e^2+8e+1}{e}$$

$$= a + (ax + e^x + e^{-x})\Big|_0^1 = e + 8 + \frac{1}{e}$$

$$= a + \left(a + e + \frac{1}{e} - 2\right) = e + \frac{1}{e} + 8$$

$$\Rightarrow \quad 2a - 2 = 8 \Rightarrow a = 5$$

2. If $I = \int_0^{\frac{\pi}{2}} \frac{\sin^{\frac{3}{2}} x}{\sin^{\frac{3}{2}} x + \cos^{\frac{3}{2}} x}\,dx$, then

$\int_0^{2I} \frac{x \sin x \cos x}{\sin^4 x + \cos^4 x}\,dx$ equals:

(1) $\frac{\pi^2}{12}$ (2) $\frac{\pi^2}{16}$

(3) $\frac{\pi^2}{4}$ (4) $\frac{\pi^2}{8}$

Answer (2)

Sol. $$I = \int_0^{\frac{\pi}{2}} \frac{(\sin x)^{\frac{3}{2}}\,dx}{(\sin x)^{\frac{3}{2}} + (\cos x)^{\frac{3}{2}}} = \int_0^{\frac{\pi}{2}} \frac{\sin^{\frac{3}{2}}\left(\frac{\pi}{2} - x\right)dx}{\sin^{\frac{3}{2}}\left(\frac{\pi}{2} - x\right) + \cos^{\frac{3}{2}}\left(\frac{\pi}{2} - x\right)}$$

$$\Rightarrow \text{ Adding } 2I = \int_0^{\frac{\pi}{2}} \frac{(\sin x)^{\frac{3}{2}} + (\cos x)^{\frac{3}{2}}}{(\sin x)^{\frac{3}{2}} + (\cos x)^{\frac{3}{2}}}\,dx = \frac{\pi}{2}$$

$$I_0 = \int_0^{\frac{\pi}{2}} \frac{x \sin x \cos x}{\sin^4 x + \cos^4 x}\,dx = \int_0^{\frac{\pi}{2}} \frac{\left(\frac{\pi}{2} - x\right)\sin x \cos x}{(\sin x)^4 + (\cos x)^4}\,dx$$

$$\text{Adding, } 2I_0 = \int_0^{\frac{\pi}{2}} \frac{\frac{\pi}{2}(\sin x)\cos x}{(\sin^4 x + \cos^4 x)}\,dx$$

$$\Rightarrow I_0 = \frac{\pi}{4}\int_0^{\frac{\pi}{2}} \frac{\tan x (\sec^2 x)\,dx}{1 + \tan^4 x}$$

put $\tan^2 x = t \Rightarrow 2\tan x \sec^2 x\,dx = dt$

$$\Rightarrow I_0 = \frac{\pi}{4}\int_0^{\infty} \frac{\frac{dt}{2}}{(1+t^2)} = \frac{\pi}{8}(\tan^{-1} t)\Big|_0^{\infty} = \frac{\pi}{8}\left(\frac{\pi}{2} - 0\right)$$

$$\Rightarrow I_0 = \frac{\pi^2}{16}$$



3. Let $x = x(y)$ be the solution of the differential equation $y = \left(x - y\frac{dx}{dy}\right)\sin\left(\frac{x}{y}\right), y > 0$ and $x(1) = \frac{\pi}{2}$.

Then $\cos(x(2))$ is equal to:

(1) $1 - 2(\log_e 2)^2$ (2) $2(\log_e 2) - 1$

(3) $1 - 2(\log_e 2)$ (4) $2(\log_e 2)^2 - 1$

Answer (4)

Sol. $y = \left(x - y\frac{dx}{dy}\right)\sin\left(\frac{x}{y}\right)$

Let $x = ty \Rightarrow \frac{dx}{dy} = y\frac{dt}{dy} + t$

$y = \left[yt - y\left(\frac{ydt}{dy} + t\right)\right]\sin t$

since $y > 0$

$1 = \left(t - \frac{ydt}{dy} - t\right)\sin t = (\sin t)\, y\left(\frac{-dt}{dy}\right)$

$\Rightarrow \frac{dy}{y} = -\sin t dt$

Integration $\int\frac{dy}{y} = \int -\sin t dt + c$

$\ln|y| = \cos t + c \Rightarrow \ln y = \cos\left(\frac{x}{y}\right) + c$

$x(1) = \frac{\pi}{2}$

$\Rightarrow \ln(1) = \cos\left(\frac{\frac{\pi}{2}}{1}\right) + c \Rightarrow c = 0$

$\Rightarrow x(y) = y\cos^{-1}(\ln y)$

$x(2) = 2\cos^{-1}(\ln 2) \Rightarrow \cos(x(2))$

$= \cos\left(2\cos^{-1}(\ln 2)\right)$

4. The distance of the line $\frac{x-2}{2} = \frac{y-6}{3} = \frac{z-3}{4}$ from the point (1, 4, 0) along the line $\frac{x}{1} = \frac{y-2}{2} = \frac{z+3}{3}$ is

(1) $\sqrt{13}$ (2) $\sqrt{15}$

(3) $\sqrt{14}$ (4) $\sqrt{17}$

Answer (3)

Sol. Line passing through (1, 4, 0) and parallel to $\frac{x}{1} = \frac{y-2}{2} = \frac{z+3}{3}$ is $L: \frac{x-1}{1} = \frac{y-4}{2} = \frac{z}{3}$

Any point on $L: (\lambda + 1, 2\lambda + 4, 3\lambda)$

Any point on $\frac{x-2}{2} = \frac{y-6}{3} = \frac{z-3}{4}$ is $(2\mu + 2, 3\mu + 6, 4\mu + 3)$

$\left.\begin{aligned}\lambda + 1 &= 2\mu + 2\\ 2\lambda + 4 &= 3\mu + 6\\ 3\lambda &= 4\mu + 3\end{aligned}\right\} \lambda = 1\ \mu = 0$

Point: (2, 6, 3)

Distance = $\sqrt{(2-1)^2 + (6-4)^2 + (3-0)^2}$

$= \sqrt{1+4+9} = \sqrt{14}$

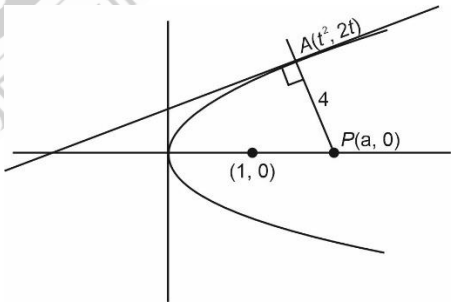
5. Let the shortest distance from $(a, 0)$, $a > 0$, to the parabola $y^2 = 4x$ be 4. Then the equation of the circle passing through the point $(a, 0)$ and the focus of the parabola, and having its centre on the axis of the parabola is

(1) $x^2 + y^2 - 6x + 5 = 0$ (2) $x^2 + y^2 - 10x + 9 = 0$

(3) $x^2 + y^2 - 4x + 3 = 0$ (4) $x^2 + y^2 - 8x + 7 = 0$

Answer (1)

Sol.



Shortest distance lie along the normal

Slope of normal at A is $(-t)$

$\Rightarrow (-t) = \frac{2t - 0}{t^2 - a}$

$\Rightarrow t = 0$ $t^2 - a = -2$

$t^2 = a - 2$



$(t^2 - a)^2 + (2t)^2 = 16$

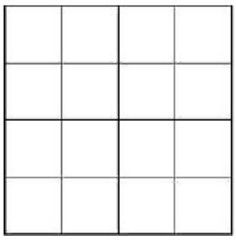
$\Rightarrow \quad t^2 = 3,\ a = 5$

Equation of circle passing through (1, 0) and (5, 0) and having centre on the axis will be diametric form

$(x - 1)(x - 5) + y.y = 0$

$x^2 + y^2 - 6x + 5 = 0$

6. A board has 16 squares as shown in the figure:



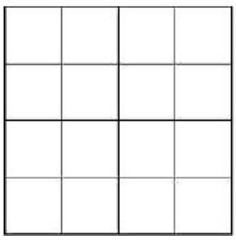
Out of these 16 squares, two squares are chosen at random. The probability that they have no side in common is

(1) $\frac{3}{5}$ (2) $\frac{23}{30}$

(3) $\frac{7}{10}$ (4) $\frac{4}{5}$

Answer (4)

Sol.



Total = $^{16}C_2$

Required ways = Total – (adjacent square)

$= {}^{16}C_2 -$

[3 pair in vertical & horizontal for each row and column]

$= {}^{16}C_2 - [3 \times 4 + 3 \times 4]$

= 96

Probability = $\frac{96}{120} = \frac{4}{5}$

7. Let $A = [a_{ij}]$ be a 3 × 3 matrix such that $A\begin{bmatrix}0\\1\\0\end{bmatrix} = \begin{bmatrix}0\\0\\1\end{bmatrix}$, $A\begin{bmatrix}4\\1\\3\end{bmatrix} = \begin{bmatrix}0\\1\\0\end{bmatrix}$ and $A\begin{bmatrix}2\\1\\2\end{bmatrix} = \begin{bmatrix}1\\0\\0\end{bmatrix}$, then a_{23} equals

(1) 1

(2) –1

(3) 2

(4) 0

Answer (2)

Sol. Let $A = \begin{bmatrix}a & b & c\\ d & e & f\\ g & h & i\end{bmatrix}$

$\therefore \quad \begin{bmatrix}a & b & c\\ d & e & f\\ g & h & i\end{bmatrix}\begin{bmatrix}0\\1\\0\end{bmatrix} = \begin{bmatrix}0\\0\\1\end{bmatrix}$

$\therefore \quad b = 0,\ e = 0,\ h = 1$

and $\begin{bmatrix}a & 0 & c\\ d & 0 & f\\ g & 1 & i\end{bmatrix}\begin{bmatrix}4\\1\\3\end{bmatrix} = \begin{bmatrix}0\\1\\0\end{bmatrix}$

$$\left.\begin{aligned}4a + 3c &= 0\\ 4d + 3f &= 1\\ 4g + 1 + 3i &= 0\end{aligned}\right\} \quad \text{...(1)}$$

and $\begin{bmatrix}a & 0 & c\\ d & 0 & f\\ g & 1 & i\end{bmatrix}\begin{bmatrix}2\\1\\2\end{bmatrix} = \begin{bmatrix}1\\0\\0\end{bmatrix}$

$$\left.\begin{aligned}2a + 2c &= 1\\ 2d + 2f &= 0\\ 2g + 1 + 2i &= 0\end{aligned}\right\} \quad \text{...(2)}$$

From equation (1) and (2) we get

$d = 1,\ f = -1$

$\therefore \quad a_{23} = -1$



8. Let the range of the function $f(x) = 6 + 16\cos x \cdot \cos\left(\frac{\pi}{3} - x\right) \cdot \cos\left(\frac{\pi}{3} + x\right) \cdot \sin 3x \cdot \cos 6x$, $x \in R$ be $[\alpha, \beta]$. Then the distance of the point (α, β) from the line $3x + 4y + 12 = 0$ is

(1) 11 (2) 9

(3) 8 (4) 10

Answer (1)

Sol. $f(x) = 6 + 16\cos x \cdot \cos\left(\frac{\pi}{3} - x\right) \cdot \cos\left(\frac{\pi}{3} + x\right) \cdot \sin 3x \cdot \cos 6x$

$f(x) = 6 + 4\cos 3x \cdot \sin 3x \cdot \cos 6x$

$\therefore$ $f(x) = 6 + \sin 12x$

$\therefore$ Range of $f(x) = [5, 7]$

$\therefore$ $[\alpha, \beta] = [5, 7]$

$\therefore$ Distance of point from $3x + 4y + 12 = 0$

$= \left|\frac{3.5 + 4.7 + 12}{\sqrt{3^2 + 4^2}}\right|$

$= 11$ units

9. A spherical chocolate ball has a layer of ice-cream of uniform thickness around it. When the thickness of the ice-cream layer is 1 cm, the ice-cream melts at the rate of 81 cm^3/min and the thickness of the ice-cream layer decreases at the rate of $\frac{1}{4\pi}$ cm/min. The surface area (in cm^2) of the chocolate ball (without the ice-cream layer) is:

(1) 256 π (2) 255 π

(3) 128 π (4) 196 π

Answer (1)

Sol. Let thickness of ice-cream is x cm at any instant and radius of chocolate ball is r cm.

$\therefore$ Volume of ice-cream $= \frac{4}{3}\pi\{(r + x)^3 - r^3\}$

$\therefore$ $V = \frac{4}{3}\pi\{(r + x)^3 - r^3\}$

$V = \frac{4}{3}\pi(3r^2x + 3rx^2 + x^3)$

On Differentiating both sides w.r.t. t we get:

$\frac{dV}{dt} = 4\pi(r^2 + 2rx + x^2) \cdot \frac{dx}{dt}$

$\therefore$ $81 = 4\pi(r^2 + 2r \cdot 1 + 1^2) \cdot \frac{1}{4\pi}$

$\therefore$ $r^2 + 2r - 80 = 0$

$\therefore$ $r = 8$.

$\therefore$ Surface area $= 4\pi r^2$

$= 4\pi \cdot 8^2$

$= 256\ \pi\ cm^2$

10. The length of the chord of the ellipse $\frac{x^2}{4} + \frac{y^2}{2} = 1$, whose mid-point is $\left(1, \frac{1}{2}\right)$, is:

(1) $\frac{1}{3}\sqrt{15}$ (2) $\frac{5}{3}\sqrt{15}$

(3) $\frac{2}{3}\sqrt{15}$ (4) $\sqrt{15}$

Answer (3)

Sol. $T = S_1$

$\frac{x \cdot 1}{4} + \frac{y}{4} = \frac{1}{4} + \frac{1}{8}$

$\Rightarrow$ $2x + 2y = 3$

$\frac{x^2}{4} + \frac{\left(\frac{3 - 2x}{2}\right)^2}{2} = 1$

$\Rightarrow$ $x = \frac{12 \pm \sqrt{120}}{12} \Rightarrow y = \frac{1}{2} \mp \frac{\sqrt{120}}{12}$

So length of chord

$= \frac{2\sqrt{15}}{3}$

11. If in the expansion of $(1 + x)^p (1 - x)^q$, the coefficients of x and x^2 are 1 and –2, respectively, then $p^2 + q^2$ is equal to :

(1) 8 (2) 18

(3) 13 (4) 20

Answer (3)



Sol. $(1+x)^p = p_{C_0} + p_{C_1}x + p_{C_2}x^2 + ... + p_{C_p}$

$(1-x)^q = q_{C_0} - q_{C_1}x + q_{C_2}x^2 ... + (-1)^q q_{C_q}$

$-q_{C_1} + p_{C_1} = 1 \Rightarrow p - q = 1$

$q_{C_2} - pq + p_{C_2} = -2$

$\frac{q(q-1)}{2} + \frac{p(p-1)}{2} - pq = -2$

$\Rightarrow \quad p = 3$

$q = 2$

$p_2 + q^2 = 13$

12. If the square of the shortest distance between the lines $\frac{x-2}{1} = \frac{y-1}{2} = \frac{z+3}{-3}$ and $\frac{x+1}{2} = \frac{y+3}{4} = \frac{z+5}{-5}$ is $\frac{m}{n}$, where m, n are coprime numbers, then $m + n$ is equal to :

(1) 21 (2) 9

(3) 6 (4) 14

Answer (2)

Sol. $a_1 = (2, 1, -3), a_2 = (-1, -3, -5)$

$$\vec{b}_1 \times \vec{b}_2 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & -3 \\ 2 & 4 & -5 \end{vmatrix}$$

$= 2\hat{i} - \hat{j}$

$$(SD)^2 = \left| \frac{(\vec{a}_2 - \vec{a}_1) \cdot (\vec{b}_1 \times \vec{b}_2)}{|\vec{b}_1 \times \vec{b}_2|} \right|^2$$

$$= \left(\frac{2}{\sqrt{5}} \right)^2 = \frac{4}{5} = \frac{m}{n}$$

$m + n = 9$

13. The number of complex numbers z, satisfying $|z| = 1$ and $\left| \frac{z}{\bar{z}} + \frac{\bar{z}}{z} \right| = 1$, is

(1) 8 (2) 10

(3) 6 (4) 4

Answer (1)

Sol. $|z| = 1$

$$\left| \frac{z}{\bar{z}} + \frac{\bar{z}}{z} \right| = 1$$

$$\Rightarrow \quad \left| z^2 + (\bar{z})^2 \right| = 1$$

Let $z = x + iy$

$\Rightarrow \quad |(x + iy)^2 + (x - iy)^2| = 1$

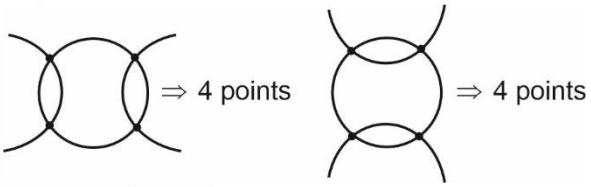
$\Rightarrow \quad |2x^2 - 2y^2| = 1$

$$\Rightarrow \quad \left| x^2 - y^2 \right| = \frac{1}{2}$$

$$\Rightarrow \quad x^2 - y^2 = \frac{\pm 1}{2}$$

and $x^2 + y^2 = 1$

Case I: $x^2 - y^2 = \frac{1}{2}$ Case II: $x^2 - y^2 = -\frac{1}{2}$



Hence, we get 8 complex number

14. A rod of length eight units moves such that its ends A and B always lie on the lines $x - y + 2 = 0$ and $y + 2 = 0$, respectively. If the locus of the point P, that divides the rod AB internally in the ratio 2 : 1 is $9(x^2 + \alpha y^2 + \beta xy + \gamma x + 28y) - 76 = 0$, then $\alpha - \beta - \gamma$ is equal to

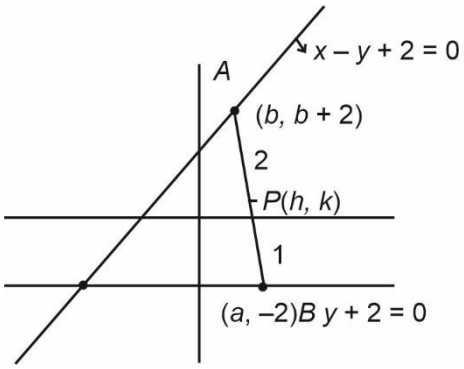
(1) 24 (2) 23

(3) 22 (4) 4

Answer (2)

Sol. $AB = 8$

$AB^2 = 64$





$\Rightarrow (a-b)^2 + (b+4)^2 = 64$ ...(1)

Now P divides AB in the ratio 2 : 1 internally

$\Rightarrow h = \frac{2a+b}{3}$ and $k = \frac{-4+b+2}{3}$

$\Rightarrow 2a + b = 3h$ ...(2) $k = \frac{b-2}{3}$

From equation (2) and (3) $\Rightarrow b = 3k + 2$ ...(3)

$\Rightarrow 2a = 3h - 3k - 2$

$\Rightarrow a = \frac{3h-3k-2}{2}$

Now by putting value of a and b in equation

$\Rightarrow \left(\frac{3h-3k-2}{2} - (3k+2)\right)^2 + (3k+2+4)^2 = 64$

$\Rightarrow \left(\frac{3h-3k-2-6k-4}{2}\right)^2 + (3k+6)^2 = 64$

$\Rightarrow (3h-9k-6)^2 + 4(3k+6)^2 = 4\times 64$

$\Rightarrow 9(h-3k-2)^2 + 36(k+2)^2 = 256$

$\Rightarrow 9(h^2 + 9k^2 + 4 - 6hk - 4h + 12k)$

$+ 36(k^2 + 4 + 4k) = 256$

$\Rightarrow 9(h^2 + 13k^2 + 20 - 6hk - 4h + 28k) = 256$

Replacing h by x and k by y

$\Rightarrow 9(x^2 + 13y^2 - 6xy - 4x + 28y) + 180 - 256 = 0$

$\Rightarrow 9(x^2 + 13y^2 - 6xy - 4x + 28y) - 76 = 0$

By comparing $\alpha = 13$, $\beta = -6$, $\gamma = -4$

$\alpha - \beta - \gamma = 13 + 6 + 4 = 23$

15. Let $A = \{(x, y) \in \mathbf{R} \times \mathbf{R}: |x + y| \geq 3\}$ and $B = \{(x, y) \in \mathbf{R} \times \mathbf{R} : |x| + |y| \leq 3\}$.

If $C = \{(x, y) \in A \cap B : x = 0 \text{ or } y = 0\}$, then

$\sum_{(x, y)\in C} |x + y|$ is

(1) 18 (2) 12

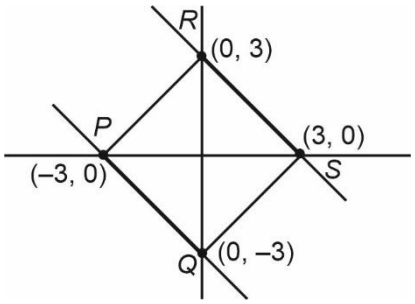
(3) 24 (4) 15

Answer (2)

Sol. $A = \{(x, y) \in \mathbf{R} \times \mathbf{R}: |x + y| \geq 3\}$

and $B = \{(x, y) \in \mathbf{R} \times \mathbf{R} : |x| + |y| \leq 3\}$

$C = \{(x, y) \in A \cap B : x = 0 \text{ or } y = 0\}$



$A \cap B$ will have only common points lying on the line PQ and RS

Now, $C = \{(-3, 0), (3, 0), (0, 3), (0, -3)\}$

$\sum_{(x, y)\in C} |x + y| = 3 + 3 + 3 + 3 = 12$

16. $\lim_{x\to\infty} \frac{(2x^2 - 3x + 5)(3x - 1)^{\frac{x}{2}}}{(3x^2 + 5x + 4)\sqrt{(3x+2)^x}}$ is equal to

(1) $\frac{2}{\sqrt{3e}}$ (2) $\frac{2e}{\sqrt{3}}$

(3) $\frac{2}{3\sqrt{e}}$ (4) $\frac{2e}{3}$

Answer (3)

Sol. $\lim_{x\to\infty} \frac{(2x^2 - 3x + 5)(3x - 1)^{x/2}}{(3x^2 + 5x + 4)\sqrt{(3x+2)^x}}$

$= \lim_{x\to\infty} \frac{x^2\left(2 - \frac{3}{x} + \frac{5}{x^2}\right)(3x)^{\frac{x}{2}}}{x^2\left(3 + \frac{5}{x} + \frac{4}{x^2}\right)(3x)^{\frac{x}{2}}} \cdot \frac{\left(1 - \frac{1}{3x}\right)^{\frac{x}{2}}}{\left(1 + \frac{2}{3x}\right)^{\frac{x}{2}}}$

$\lim_{x\to\infty}\left(1 - \frac{1}{3x}\right)^{\frac{x}{2}} = e^{\lim_{x\to\infty}\left(1-\frac{1}{3x}-1\right)\times\frac{x}{2}} = e^{\frac{-1}{6}}$

$\lim_{x\to\infty}\left(1 + \frac{2}{3x}\right) = e^{\lim_{x\to\infty}\left(1+\frac{2}{3x}-1\right)\times\frac{x}{2}}$

$= e^{\frac{1}{3}}$

So, $\frac{2}{3}\times\frac{e^{\frac{-1}{6}}}{e^{\frac{1}{3}}} = \frac{2}{3}\times\frac{1}{e^{\frac{1}{3}+\frac{1}{6}}} = \frac{2}{3\sqrt{e}}$



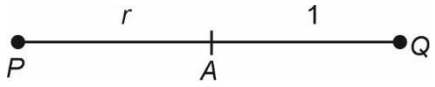
17. Let the point A divide the line segment joining the points $P(-1, -1, 2)$ and $Q(5, 5, 10)$ internally in the ratio $r : 1 (r > 0)$. If O is the origin and $\left(\overrightarrow{OQ}\cdot\overrightarrow{OA}\right) - \frac{1}{5}\left|\overrightarrow{OP}\times\overrightarrow{OA}\right|^2 = 10$, then the value of r is:

(1) $\sqrt{7}$ (2) 7

(3) 3 (4) 14

Answer (2)

Sol.



$$A = \left(\frac{5r-1}{r+1}, \frac{5r-1}{r+1}, \frac{10r+2}{r+1}\right)$$

$$\left(\overrightarrow{OQ}\cdot\overrightarrow{OA}\right) - \frac{1}{5}\left|\overrightarrow{OP}\times\overrightarrow{OA}\right|^2 = 10$$

$$\overrightarrow{OQ} = 5\hat{i} + 5\hat{j} + 10\hat{k}$$

$$\overrightarrow{OA} = \frac{5r-1}{r+1}\hat{i} + \frac{5r-1}{r+1}\hat{j} + \frac{10r+2}{r+1}\hat{k}$$

$$\overrightarrow{OP} = -\hat{i} - \hat{j} + 2\hat{k}$$

$$\overrightarrow{OP}\times\overrightarrow{OA} = \frac{1}{r+1}\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 5r-1 & 5r-1 & 10r+2 \\ -1 & -1 & 2 \end{vmatrix}$$

$$= \frac{1}{r+1}(\hat{i}(20r) - \hat{j}(20r))$$

$$= 5\left(\frac{5r-1}{r+1}\right) + 5\left(\frac{5r-1}{r+1}\right) + 10\left(\frac{10r+2}{r+1}\right)$$

$$-\frac{1}{5}\left(\frac{2\times 400r^2}{(r+1)^2}\right) = 10$$

$$\frac{150r+10}{r+1} - \frac{1}{5}\left(\frac{2\times 400r^2}{(r+1)^2}\right) = 10$$

$(150r + 10)(r + 1) - 160r^2 = 10(r + 1)^2$

$(15r + 1)(r + 1) - 16r^2 = (r + 1)^2$

$15r^2 + 16r + 1 - 16r^2 = r^2 + 2r + 1$

$-2r^2 + 14r = 0$

$r = 0, 7$

18. The system of equations

$x + y + z = 6$,

$x + 2y + 5z = 9$,

$x + 5y + \lambda z = \mu$

has no solution if

(1) $\lambda \neq 17, \mu \neq 18$ (2) $\lambda = 17, \mu = 18$

(3) $\lambda = 15, \mu \neq 17$ (4) $\lambda = 17, \mu \neq 18$

Answer (4)

Sol. $\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 1 & 2 & 5 & 9 \\ 1 & 5 & \lambda & \mu \end{array}\right] \sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 4 & 3 \\ 0 & 4 & \lambda-1 & \mu-6 \end{array}\right] \sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 4 & 3 \\ 0 & 0 & \lambda-17 & \mu-18 \end{array}\right]$

$\lambda = 17$ and $\mu \neq 18$

19. Let $\int x^3 \sin x dx = g(x) + C$, where C is the constant of integration. If $8\left(g\left(\frac{\pi}{2}\right) + g'\left(\frac{\pi}{2}\right)\right) = \alpha\pi^3 + \beta\pi^2 + \gamma$, $\alpha, \beta, \gamma \in Z$, then $\alpha + \beta - \gamma$ equals:

(1) 62

(2) 55

(3) 48

(4) 47

Answer (2)

Sol. $g(x) + C = \int x^3 \cdot \sin x dx$

$= -x^3\cos x + 3x^2\sin x$

$+6x\cos x - 6\sin x + C$

Hence, $g(x) = -x^3\cos x + 3x^2\sin x + 6x\cos x - 6\sin x$

$$g\left(\pi/2\right) = \frac{3\pi^2}{4} - 6$$

Also, $g'(x) = x^3 \sin x \Rightarrow g'\left(\pi/2\right) = \frac{\pi^3}{8}$

$$8\left(g\left(\pi/2\right) + g'\left(\pi/2\right)\right) = 8\left(\frac{3\pi^2}{4} - 6 + \frac{\pi^3}{8}\right)$$

$= 6\pi^2 + \pi^3 - 48 = \alpha\pi^3 + \beta\pi^2 + \gamma$

$\Rightarrow \alpha = 1, \beta = 6, \gamma = -48$

Hence, $\alpha + \beta - \gamma = 1 + 6 + 48 = 55$



20. Let $X = R \times R$. Define a relation R on X as:

$$(a_1, b_1)\ R\ (a_2, b_2) \Leftrightarrow b_1 = b_2$$

Statement I : R is an equivalence relation.

Statement II: For some $(a, b) \in X$, the set $S = \{(x, y) \in X : (x, y)\ R\ (a, b)\}$ represents a line parallel to $y = x$.

In the light of the above statements, choose the correct answer from the options given below:

(1) Statement I is true but Statement II is false

(2) Statement I is false but Statement II is true

(3) Both Statement I and Statement II are false

(4) Both Statement I and Statement II are true

Answer (1)

Sol. $(a_1, b_1)R(a_2, b_2) \Leftrightarrow b_1 = b_2$

Check reflexive

$(a_1, b_1)R(a_1, b_1)$

$\Rightarrow\ b_1 = b_1$

$\Rightarrow$ Reflexive

Check symmetric

$(a_1, b_1)R(a_2, b_2) \Leftrightarrow b_1 = b_2$

$\Rightarrow (a_2, b_2)R(a_1, b_1) \Leftrightarrow b_2 = b_1$

$\Rightarrow\ b_1 = b_2 \Rightarrow$ symmetric

Check transitive

$\Rightarrow (a_1, b_1)R(a_2, b_2) \Leftrightarrow b_1 = b_2$

$(a_2, b_2)R(a_3, b_3) \Leftrightarrow b_2 = b_3$

$\Rightarrow\ b_1 = b_2 = b_3$

$\Rightarrow\ b_1 = b_3$

$\Rightarrow\ (a_1, b_1)\ R(a_3, b_3)$

$\Rightarrow\ R$ is transitive

$\Rightarrow\ R$ is equivalence

II :

$(x, y)R(a, b) \Leftrightarrow y = b \neq y = x$

$\Rightarrow$ Not true.

SECTION - B

Numerical Value Type Questions: This section contains 5 Numerical based questions. The answer to each question should be rounded-off to the nearest integer.

21. The roots of the quadratic equation $3x^2 - px + q = 0$ are 10th and 11th terms of an arithmetic progression with common difference $\frac{3}{2}$. If the sum of the first 11 terms of this arithmetic progression is 88, then $q - 2p$ is equal to _____.

Answer (474)

Sol. $S_{11} = \frac{11}{2}[2a + 10d] = 88$

$a + 5d = 8$

$\Rightarrow\ a = \frac{1}{2}$

$T_{10} + T_{11} = \frac{p}{3} = a + 9d + a + 10d$

$\Rightarrow \frac{p}{3} = \frac{59}{2} \Rightarrow p = \frac{177}{2}$

$T_{10} \cdot T_{11} = \frac{q}{3}$

$(a + 9d)(a + 10d) = \frac{q}{3}$

$\left(\frac{1}{2} + 9 \times \frac{3}{2}\right)\left(\frac{1}{2} + 10 \times \frac{3}{2}\right) = \frac{q}{3}$

$\Rightarrow\ q = 651$

$q - 2p = 651 - 177 = 474$

22. The variance of the numbers 8, 21, 34, 47, …, 320 is _____.

Answer (8788)

Sol. 8, 21, 34 ….320

$320 = 8 + (n - 1)\ 13$

$\Rightarrow\ n = 25$



$$\text{Mean} = \frac{\sum x_i}{n} = \frac{8+21+34+...+320}{25} = \frac{\frac{25}{2}[8+320]}{25}$$

$= 164$

$$\text{Variance} = \frac{\sum x_i^2}{n} - [\text{mean}]^2$$

$$= \frac{8^2+21^2+...+320^2}{25} - (164)^2$$

$= 8788$

23. The focus of the parabola $y^2 = 4x + 16$ is the centre of the circle C of radius 5. If the values of λ, for which C passes through the point of intersection of the lines $3x - y = 0$ and $x + \lambda y = 4$, are λ_1 and λ_2, $\lambda_1 < \lambda_2$, then $12\lambda_1 + 29\lambda_2$ is equal to _____.

Answer (15)

Sol. $y^2 = 4(x + 4)$

Equation of circle

$(x + 3)^2 + y^2 = 25$

It passes through the point of intersection of two lines $3x - y = 0$ and $x + \lambda y = 4$

$\left(\frac{4}{3\lambda+1}, \frac{12}{3\lambda+1}\right)$, we get

$$\lambda = \frac{-7}{6}, 1; 12\lambda_1 + 29\lambda_2 = -14 + 29 = 15$$

24. Let α, β be the roots of the equation $x^2 - ax - b = 0$ with $\text{Im}(\alpha) < \text{Im}(\beta)$. Let $P_n = \alpha^n - \beta^n$. If $P_3 = -5\sqrt{7}i$, $P_4 = -3\sqrt{7}i$, $P_5 = 11\sqrt{7}i$ and $P_6 = 45\sqrt{7}i$, then $|\alpha^4 + \beta^4|$ is equal to _____.

Answer (31)

Sol. $\alpha + \beta = a$, $\alpha\beta = -b$

$P_6 = aP_5 + bP_4$

$45\sqrt{7}i = a \times 11\sqrt{7}i + b(-3\sqrt{7})i$

$45 = 11a - 3b$ ...(i)

and $P_5 = aP_4 + bP_3$

$11\sqrt{7}i = a\left(-3\sqrt{7}i\right) + b(-5\sqrt{7}i)$

$11 = -3a - 5b$ ...(ii)

$a = 3$, $b = -4$

$$|\alpha^4 + \beta^4| = \sqrt{(\alpha^4 - \beta^4)^2 + 4\alpha^4\beta^4}$$

$$= \sqrt{-63 + 4.4^4}$$

$$= \sqrt{-63 + 1024}$$

$$= \sqrt{961}$$

$= 31$

25. The number of ways, 5 boys and 4 girls can sit in a row so that either all the boys sit together or no two boys sit together, is_____.

Answer (17280)

Sol. Case I, when all boys sit together

$B_1B_2B_3B_4B_5\ G_1G_2G_3G_4$

$= 5!\ 5! = 14400$

Case II, when no two boys sit together

$\uparrow G_1 \uparrow G_2 \uparrow G_3 \uparrow G_4 \uparrow$

$4! \cdot 5! \Rightarrow 2880$

$\therefore$ Total number of ways = 14400 + 2880

= 17280



PHYSICS

SECTION - A

Multiple Choice Questions: This section contains 20 multiple choice questions. Each question has 4 choices (1), (2), (3) and (4), out of which **ONLY ONE** is correct.

Choose the correct answer :

26. A circular disk of radius R meter and mass M kg is rotating around the axis perpendicular to the disk. An external torque is applied to the disk such that $\theta(t) = 5t^2 - 8t$, where $\theta(t)$ is the angular position of the rotating disc as a function of time t.

How much power is delivered by the applied torque, when $t = 2$ s?

(1) $72MR^2$ (2) $8MR^2$

(3) $108MR^2$ (4) $60MR^2$

Answer (4)

Sol. $I = \frac{MR^2}{2}$

$\omega = 10t - 8 = 12$ for $t = 2$

$\alpha = 10$

$P = \tau\omega = \frac{MR^2}{2} \times 10 \times 12$

$= 60MR^2$

27. A plane electromagnetic wave of frequency 20 MHz travels in free space along the $+x$ direction. At a particular point in space and time, the electric field vector of the wave is $E_y = 9.3\ \text{Vm}^{-1}$. Then, the magnetic field vector of the wave at that point is

(1) $B_Z = 1.55 \times 10^{-8}$ T

(2) $B_Z = 6.2 \times 10^{-8}$ T

(3) $B_Z = 9.3 \times 10^{-8}$ T

(4) $B_Z = 3.1 \times 10^{-8}$ T

Answer (4)

Sol. $E = CB$

$B = \frac{9.3}{3 \times 10^8}$

$= 3.1 \times 10^{-8}$

28. The width of one of the two slits in Young's double slit experiment is d while that of the other slit is xd. If the ratio of the maximum to the minimum intensity in the interference pattern on the screen is 9 : 4 then what is the value of x?

(Assume that the field strength varies according to the slit width.)

(1) 4 (2) 5

(3) 3 (4) 2

Answer (2)

Sol. $A \propto$ width of slit

$\Rightarrow A_1 = A, A_2 = xA$

$\frac{I_{max}}{I_{min}} = \frac{(A_1 + A_2)^2}{(A_1 - A_2)^2} = \frac{(x+1)^2}{(x-1)^2} = \frac{9}{4}$

$2x + 2 = 3x - 3$

$x = 5$

29. The energy of a system is given as $E(t) = \alpha^3 e^{-\beta t}$, where t is the time and $\beta = 0.3\ \text{s}^{-1}$. The errors in the measurement of α and t are 1.2% and 1.6%, respectively. At $t = 5$ s, maximum percentage error in the energy is :

(1) 6%

(2) 4%

(3) 8.4%

(4) 11.6%

Answer (1)

